

SKILL – 10

PART 1 — Current Directory, Exit Request and Resource Cleanup

Code:

```
GNU nano 2.7 skill10_directory_exit.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <string.h>

int main()
{
    char cwd[1024];

    printf("=== SKILL 10: DIRECTORY AND EXIT HANDLING ===\n");

    /* Retrieve current working directory */
    if (getcwd(cwd, sizeof(cwd)) != NULL)
    {
        printf("Current Directory: %s\n", cwd);
    }
    else
    {
        perror("getcwd");
        return 1;
    }

    /* Display path */
    printf("Directory Path: %s\n", cwd);

    /* Save state */
    FILE *fp = fopen("skill10_state.txt", "w");
    if (fp == NULL)
    {
        perror("File open failed");
        return 1;
    }

    fprintf(fp, "Saved Directory: %s\n", cwd);
    fprintf(fp, "Program State: Active\n");
}
```

Output:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-10$ ./skill10_directory_exit
=== SKILL 10: DIRECTORY AND EXIT HANDLING ===
Current Directory: /home/bhardwaj/2520090182_OSSP/2520090182_Skill/Skill-10
Directory Path: /home/bhardwaj/2520090182_OSSP/2520090182_Skill/Skill-10
State saved successfully.
Resources cleaned up successfully.
Process exiting successfully...
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-10$ |
```

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-10$ ./skill10_directory_exit
=== SKILL 10: DIRECTORY AND EXIT HANDLING ===
Current Directory: /home/bhardwaj/2520090182_OSSP/2520090182_Skill/Skill-10
Directory Path: /home/bhardwaj/2520090182_OSSP/2520090182_Skill/Skill-10
State saved successfully.
Resources cleaned up successfully.
Process exiting successfully...
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-10$ cat skill10_state.txt
Saved Directory: /home/bhardwaj/2520090182_OSSP/2520090182_Skill/Skill-10
Program State: Active
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-10$ |
```

PART 2 — EXPORT Environment Variable

Code:

```
GNU nano 2.7 skill10_export.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>
#include <unistd.h>
#include <sys/wait.h>

int valid_name(const char *name)
{
    int i;

    if (name[0] == '\0')
        return 0;

    if (!isalpha(name[0]) && name[0] != '_' )
        return 0;

    for (i = 1; name[i] != '\0'; i++)
    {
        if (!isalnum(name[i]) && name[i] != '_' )
            return 0;
    }

    return 1;
}

int main()
{
    char input[200];
    char *name;
    char *value;
    char *equal;
    pid_t pid;

    printf("=== SKILL 10: EXPORT ENVIRONMENT VARIABLE ===\n");
    printf("Enter command: ");
}
```

Output:

```

bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-10$ ./skill10_export
=== SKILL 10: EXPORT ENVIRONMENT VARIABLE ===
Enter command: export MYNAME=Bhardwaj
Variable Name: MYNAME
Variable Value: Bhardwaj
New variable. Creating...
Environment variable updated successfully.
Current Value: Bhardwaj

--- Child Process ---
Child received MYNAME=Bhardwaj
Testing variable using printenv:
Bhardwaj

Parent process: Child completed successfully.

```

Test existing variable/update

```

bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-10$ ./skill10_export
=== SKILL 10: EXPORT ENVIRONMENT VARIABLE ===
Enter command: export MYNAME=Surya
Variable Name: MYNAME
Variable Value: Surya
New variable. Creating...
Environment variable updated successfully.
Current Value: Surya

--- Child Process ---
Child received MYNAME=Surya
Testing variable using printenv:
Surya

Parent process: Child completed successfully.

```

Test invalid variable name

```

Parent process: Child completed successfully.
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-10$ nano skill10_export.c
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-10$ ./skill10_export
=== SKILL 10: EXPORT ENVIRONMENT VARIABLE ===
Enter command: export 123ABC=hello
Error: Invalid variable name '123ABC'
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-10$ |

```

Test invalid syntax

```

bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-10$ ./skill10_export
=== SKILL 10: EXPORT ENVIRONMENT VARIABLE ===
Enter command: set MYNAME=hello
Error: Invalid export syntax.
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-10$ |

```